

Greece

Hellenic Competition Commission

I. Introduction

The Hellenic Competition Commission ('HCC') has been one of the first movers, among competition authorities in Europe, in the study and integration of advanced computational competition law and economics tools in its day-to-day activities.²⁶ Early on taking over the Organization, in 2020, the new Agency leadership realized the importance of AI and data science tools in enhancing the pace and scope of competition law investigations and put in place a holistic strategic vision with the aim to transform the HCC to an AI-augmented competition authority by 2024.

In pursuing this overall ambition, the HCC first completed in 2020 a comprehensive inception report with the aim to map the use of different computational techniques (Big Data, AI, machine learning, deep learning) by competition authorities around the world, focusing in particular on some selected competition authorities, in view of their different level of development, size and legal system in order to better understand the linkage between the use of these computational techniques and institutional change (HCC, Inception Report, Computational Competition Law and Economics, 2020).²⁷ In the meantime, the HCC invested in the construction of a bespoke data infrastructure for the collection and analysis of data and economic intelligence, as well as proceeded to the procurement of number of commercial off-the-shelf AI software products (*e.g.*, Tovek Intelligence) for use by the newly formed forensic and market mapping units of the HCC (2020).

The HCC Data Analytics and Economic Intelligence Platform (DAECI) was conceived of as a tool to integrate and keep updated multiple external data sources in common database schema with the aim to provide visualization tools for data exploration and a screening device to HCC's economists and lawyers. The platform is being hosted within the premises of the HCC to the users having the appropriate access credentials. The application is being deployed as a Flask Server that incorporates a number of dashboards that enable various forms of data visualization

²⁶ Written by Ioannis Lianos, Vassilis Vassalos, Petros Boulrieris, Maria Niki Fourka, Nikos Delis, Xanthos Xanthopoulos, Lampis Tzai.

²⁷ Hellenic Competition Commission, *Computational Competition Law and Economics Inception Report*, <https://www.epant.gr/en/enimerosi/publications/research-publications/item/1414-computational-competition-law-and-economics-inception-report.html> (last accessed June 7, 2023).

(e.g., box plots, basket plots, data error bars). The Platform also integrates a screening method to detect anti-competitive practices—including cartels and excessive pricing— from the analysis of market data (in particular prices, but also for some products quantities), taking advantage of new legislation enabling the authority to have mandated access to primary data regarding prices by the main supermarkets in the country, the distribution system for petrol stations, and the Athens central market for vegetables and fruits. This enables the authority to follow daily the level of prices for 2200 product codes across the country and to be able to use a time series since January 2020 and for some products a few years earlier. The interactive dashboards are created by utilizing the framework of *Plotly-Dash* that is known for its usability features and scalability.²⁸ The screening tool was implemented in number of investigations opened by the Hellenic Competition Commission since 2020. Moreover, the Commission is investing in its expandable Big Data Management Infrastructure Platform/ dashboard, for the collection of real-time public data from different sources (Price Observatory of Supermarkets, fuel prices, vegetables and fruits prices, public procurement data, etc.), data that is automatically uploaded and updated every day or many times per week. Furthermore, the HCC has recently (2023) proceeded to cooperation with experts to design a program, drawing raw data from unstructured information available in the national public procurement database and other sources. This data will be mainly used for cartel-detection but will also offer an integrated data analytics environment with various tools/apps, on the basis of bespoke programs and /or available off the shelf software tools to visualize and analyze data.

In order to successfully complete the above-mentioned holistic strategy, the HCC completed in April 2021 the recruitment process of a chief technology officer and a team of data-scientists who joined the HCC for a renewable mandate of two years. The team is headed by Professor Vassilis Vassalos, professor of informatics at the Athens Economic University who is the Chief Data Scientist of the HCC, three data scientists (Petros Boulteris, Maria Niki Fourka, Lampis Tzai) and two computer scientists (Nikos Delis, Xanthos Xanthopoulos). The team has been working on several programs requested by HCC's management with the aim to transforming all aspects of the work of the HCC. The projects undertaken relate (i) to the development of data analysis tools to facilitate HCC's staff in mapping and monitoring daily markets, (ii) the construction of bespoke algorithms for the purposes of HCC's investigations and more generally the implementation of machine learning and data analysis models to reduce the time required to analyze cases, since HCC employees will be able to take on and process more cases at the same time, (iii) the development of reusable code to

²⁸ Plotly, Dash Enterprise, <https://plotly.com/dash/> (last accessed June 7, 2023).

accelerate common workloads, (iv) the establishment of a project management platform so as to enable the day-to-day monitoring of the work of the HCC's employees with the aim to enhance more flexible forms of working (teleworking) and administrative efficiency.

This brief note will focus on some of the computational competition law and economics applications of the HCC in different work streams. In particular, the Data Science team at HCC has contributed in (i) mapping markets and harvesting data from supermarkets/groceries through the development of the DAECI platform, (ii) providing data analysis for the ongoing fuels market investigation, (iii) assisting the HCC in the monitoring of the household basket data tool put in place by the Ministry of Development, (iv) engaging in the technology-assisted review of data in the inter-bank collusion investigation.

Furthermore, we will succinctly present some of the ongoing workstreams of the data science team, in particular (i) the implementation of an AI-enabled document library to enable semantically similar file retrieval, (ii) developing tools for social media and company monitoring, and (iii) proceeding to an AI-revamp of the HCC's existing whistle-blower platforms.

II. Existing workstreams in the work of the data science team

A. The HCC Data Analysis and Economic Intelligence Platform ('DAECI')

1. The construction of the DAECI

When conducting legal investigations, the Electronic Discovery Reference Model is a comprehensive approach to digital data. The EDRM has 9 basic stages: Information Governance, Identification, Preservation, Collection, Processing, Review, Analysis, Production and Presentation. The HCC is increasingly faced with digital data when conducting legal investigations, requiring a significant investment of time in data collection, processing, and analysis, so streamlining more common and routine tasks is important.

The purpose of the HCC Data Analytics and Economic Intelligence (DAECI) platform is to create an easy-to-use data analysis environment, with the aim of collecting market data from publicly available databases in real time, storing them in a local database, editing, visualizing and analyzing them, as well as the development

of screening tools using Machine Learning in order to detect anti-competitive practices (market screens) in various sectors of the Greek market.

The main sources for data collection so far (in the first year of operation of the DAECI platform) are e-katanalotis (Market Observatory) for monitoring food prices as well as the basic products of the Greek household, OKAA (Central Markets and Fisheries Organization) and Eurostat for monitoring fruit prices, vegetables, meats and fish, as well as fuelprices.gr for fuel prices in the Greek market (See Table 1).

	Supermarkets	Fuels	Fruits and Vegetables	Fish
Brand/Sale Region	16 Brands	20 Brands	Wholesale	10 regions
Categories	123	-	3	5
Products	>1200	105	350	148
Years	2014 - 2016 2020 - 2023	2015 - 2018 2020 - 2023	2017 - 2023	2018 - 2023
Data Frequency	Daily	Daily	1-3 times per week	Monthly
Type of Prices	Shelf price	Offering Prices	Offering Prices	Min-Mean-Max of selling price
Quantity		-	-	Yes

Table I: Price monitoring

The platform uses existing APIs to extract the information it needs, but possesses the extensibility to manage other types of data that can help the work of the Competition Commission such as standalone files (e.g., .csv, .xlsx, .pkl) and/or direct links to other databases within the Commission. After data collection there is always the possibility of errors during data entry, especially if done by a human. For this reason, input data are processed in order to remove anomalies such as gaps at various intervals, inconsistent data types (typographical errors, etc.) or incorrect categorization. The platform, to solve the above, undertakes the creation of new product categories and their integration into its database using Natural Language Processing (NLP).

The DAECI platform provides the user with specially designed dashboards for market monitoring so that quantitative analysis can be performed immediately, and reporting can be made easier. The user can select the type of visualization from various types of diagrams (box plot, time series diagram, deviation range plot, custom basket plot) and at the same time make changes to its parameters to make it easier to explore the data during modeling time. In more detail, the time series chart identifies price trends for different products and categories, for one or more companies. The graph gives a more detailed picture of price changes by category or product using measures such as quartiles, median, safety margin and extremes. The deviation range diagram describes the minimum, average and maximum selling price of catches per month, for each month.

To create the custom basket diagram, a default basket was created, indicative of the average Greek household, with 16 popular categories based on the analysis of the Consumer Goods Research Institute (IELKA), weighing the product categories with the 2019 Consumer Price Index Authority (ELSTAT). The above diagrams give the possibility to select different time units (daily, weekly, day of the week, monthly), selection of the appropriate meter per case (average, median), observation of values with or without seasonality, definition of the desired monitoring period, descriptive and statistics for selected quantities, data mining and direct comparisons between charts.

For the analysis of the data with Machine Learning algorithms, the experience and knowledge of HCC experts were used both for the design stage and for the implementation stage. So far, we have created a two-step screening mechanism to allow fast data-driven prioritization of Commission cases. Upon selection of a time period and some potentially suspicious products to investigate, the screening mechanism produces various relevant metrics (e.g., T-tests, Log-Differences in

Differences) and a diagram to help the user assess whether the counterfactuals show unusual and suspicious changes in the prices of the products under consideration. Finally, the mechanism calculates the expected value of the suspicious product, and it is identified if there is a structural break in the time series. All of the above bring the Commission one step closer to its original goal which is none other than mapping the Greek market and having intelligent tools at its disposal to monitor the state of competition among firms.

2. Completing the DAECI with data harvested from e-shops

Furthermore, the HCC Data Science team has recently completed the DAECI with a more extensive and systematic data harvesting from e-shops with the aim to constitute a complete catalog of every product offered by supermarkets that operate in Greece and have an e-shop and the constitution of time series, starting in 2023. The data collected take the form of JSON files for every product 61300 products in total as of this writing) from every supermarket and includes the name of each product, the prices daily and the SKUs/Barcodes. The data also includes discounts, eventually promotion campaigns and offer stickers, when applicable.

The custom tools elaborated by the HCC Data Science team harvest automatically information from each e-shop daily and raw data is backed up before any preprocessing. Following this stage, the data is processed and inserted in the database. With this new tool the HCC stays up-to-date on the latest pricing & product information and can implement the structural breaks methodology that has been put in place for the DAECI platform at a larger scale, without also being dependent on access to the e-katanalotis app, which constitutes the main source of information on which relies the DECI. The project can be easily scaled up to include additional supermarkets. Other e-commerce platforms can be added in the future, and automating the data collection process would allow the HCC to save time & resources.

B. Providing data analytics assistance to the Fuels Market Investigation

Systematically using data analytics tools may largely enhance the capacity of the competition authority to complete more timely sector inquiries and market investigation references, in particular, if there are large data sets available.

One of the projects on which the HCC Data Science worked team in 2022 and 2023 concerned the preparation and processing of data for the Fuels markets mapping and Market Investigation Reference in 2022 and 2023. The team proceeded

to data collection about prices for the period Q3 2019 - Q3 2022 regarding different levels of the fuels supply chain:

- Price information for crude oil;
- Invoices from refineries to major wholesalers (Refiner $\rightarrow$ W);
- Invoices from wholesalers to gas stations (W $\rightarrow$ Retail);
- Store prices for gasoline, diesel, and heating oil at every gas station in Greece (Retail $\rightarrow$ Public).

While the estimated time to process data by hand was initially 2 months with the involvement of 4 employees, the HCC Data Science team was able to complete this task in 3 weeks, with the involvement of 2 employees. Data harvesting was complicated by the fact that not all information on prices was publicly available:

- There was price information for crude oil, although the team employed price imputation for days where the Brent & WTI Barrel prices were not available.
- With regard to invoices from refineries to major wholesalers (Refiner $\rightarrow$ W), most discounts were applied directly on the invoice, while others were granted through "Credit Invoices" that had negative values and were used to give discounts or bonuses to customers. The team also had to process the invoices to obtain the actual, final price per liter, taking into account discounts, credit invoices and VAT. The team matched the wholesaler names on the refiner's invoices to the wholesaler invoices using the wholesaler name as a unique identifier.
- With regard to invoices from wholesalers to gas stations (W $\rightarrow$ Retail), as before, the team had to process the invoices to obtain the actual, final price per liter, taking into account discounts, credit invoices, and VAT. It also matched the retailer names on the wholesaler's invoices to the retailer store prices using the retailer's name as a unique identifier. Note that many retailers had undergone a change of management, and the station name column changed as a result. This created a problem as multiple gas stations could share the same physical address, which would be impossible. The team used fuzzy string matching at first, then NLP to ensure that each station address uniquely described a station.
- With regard to store prices for gasoline, diesel, and heating oil at every gas station in Greece (Retail $\rightarrow$ Public), the Data Science team collected information at the station-service level from the public database `fuelprices.gr` which is station-reported and covers the whole country,

although it also proceeded to rectify some errors in the database as many vital pieces of it such as addresses, station names, and postal codes were incorrect or missing. The Google Maps geocoding API was employed to solve the issue.

C. Monitoring the ‘Household Basket’

In the context of its anti-inflationary policies, the Greek Ministry of Development established the ‘Household Basket Initiative’ with the aim to increase competition among retail chains for categories of ‘basic consumer goods’ and therefore to limit the likelihood for sudden and steep price hikes. According to this initiative, food and other product hypermarkets that carry out retail activities (supermarkets), the total annual turnover of which exceeds ninety million (90,000,000) euros per year, based on the financial statements of the previous business year, are required to send to the Ministry at regular intervals, a list of consumer products (‘household basket’) that are necessary for a decent living of households and are available at affordable prices. The list of the products included in the ‘household basket’ for each party liable and their prices is set freely by the undertakings and products included in the ‘household basket’ are checked as a matter of priority by the competent supervisory authorities for compliance with the provisions on unfair profiteering. The products in the list may be differentiated each time a new list is sent to the Ministry of Development and Investments. Those liable list at least one product from each product category included in the household basket and ensure that it is available at an affordable price, compared to other products of the same category. They also ensure that the consumer, who comes to their physical stores or accesses their digital marketplaces, is informed about the composition of the ‘household basket’ that they propose by any appropriate means such as product catalogs, brochures and specific labeling on the products or at the marketplaces. The non-sending of a product catalog or the incomplete sending of a product catalog by the obligors shall incur a pecuniary administrative penalty of five thousand (5,000) euros for each day of delay in sending a complete catalog.

The HCC expressed the Opinion that the regulatory initiative does not raise concerns from a competition law perspective and, in any case, despite the distortions of competition that may be caused by this measure, it is proportionate, in view of the current difficult circumstances for consumers and given the temporary nature of the measure. However, the HCC also raised concerns with regard to the fact that that supermarkets selecting the products they wish to include in the household basket may offer them a leverage capacity in negotiations with suppliers of such products,

especially with those with a weak bargaining power, or that such initiative may lead to systematic self-preferencing practices by supermarkets for their own private label products, against branded products, while it cannot be excluded that there could be price increases on products or product categories (from the same or different suppliers) off-basket to compensate for any losses from making the in-basket products available at affordable (or possibly reduced) prices. The HCC also expressed its intention to monitor the implementation of the ‘household basket.’

The HCC Data Science team was in charge of developing analytic tools that would facilitate the monitoring of the operation of the ‘household basket’ and controlling for anti-competitive behavior adopted by supermarkets, thus limiting the time and human resources necessary for performing this task. Thus, it put in place a useful and up-to-date dashboard that analyzes household basket data. Using this tool, the prices charged by the stores are monitored, enabling analysts to recognize abnormal trends in the pricing of goods. Clicking on each supermarket in the weekly basket total cost chart breaks down the basket structure for the selected supermarket. The basket structure chart shows the contribution of each category to the total price of the basket.

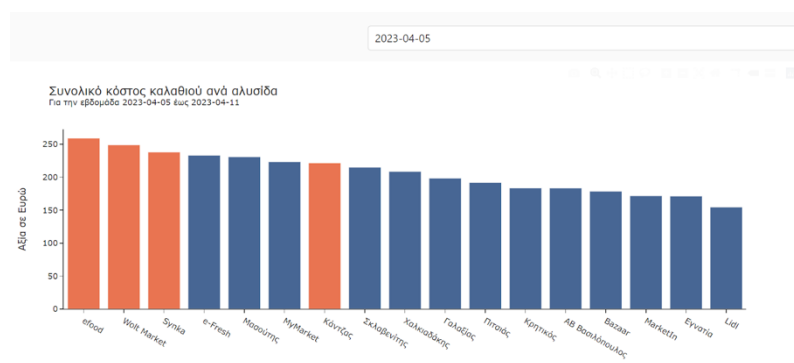


Figure I: Household basket

The HCC Data Science team also developed *between-category time series* by retailers, which shows the average price of product categories for each supermarket over the selected time period. Each supermarket is represented by a different colored line in the diagram. The price of each line represents the average price of products in that category in that supermarket. This chart can help analysts monitor product price trends by category in supermarkets and discover any price anomalies that may indicate anti-competitive behavior by supermarkets. Finally, the team also developed the *within-category time series by the retailer*, which shows the evolution of the price of all products belonging to a selected category for each supermarket. Each

line in the chart corresponds to a specific product and retailer and is colored differently for easier identification.

D. Technology-assisted review of data in the inter-bank collusion investigation

The HCC Data Science team has also worked on tools facilitating the technology-assisted review of data, in particular in the ongoing inter-bank collusion investigation, some of which are briefly examined below.

1. Attachment Analysis

In the Hellenic Competition Commission's (HCC) investigation, attachment analysis played a crucial role in filtering out irrelevant emails and identifying attachments that required further review.

To extract the text from each attachment, the data science team utilized Apache Tika, a powerful text extraction library. This enabled the extraction of text from various file formats, including .doc, .pdf, .xls, and more. By converting the attachments into machine-readable text, the team could perform further analysis and processing. To ensure the quality and accuracy of the extracted text, the team employed natural language processing (NLP) methods. Techniques such as *Word2Vec*, *SpaCy*, and *GreekBERT* were used for text cleaning, normalization, and semantic analysis. Text embeddings were configured to represent the extracted text in a numerical format suitable for machine learning algorithms. These embeddings captured the semantic meaning and context of the text, allowing for effective comparison and similarity analysis.

For each email flagged as important, pairwise similarity scores were calculated between the attachments of crucial emails and other emails using the configured text embeddings. This process identified attachments that exhibited high similarity to the crucial emails. The five most similar attachments were selected for further review. Attachments belonging to unread emails were given priority, as they were likely to contain crucial information not yet analyzed. These attachments were returned and suggested for manual inspection by HCC employees, reducing the number of attachments requiring review and focusing on potentially significant evidence.

2. Graph Analysis of Corporate Entities in Emails

Graph analysis was employed to analyze the corporate entities involved in the email communication. For each email, the To/From/CC/BCC fields were extracted to identify the members involved in the email thread. Entity analysis techniques were applied to accurately identify and classify the entities. The analysis also detected aliases and variations for each entity, enhancing the accuracy of subsequent analysis steps. The entities and emails were represented as nodes and edges, respectively, in a graph structure. Each email thread connected two or more nodes, creating a network representation of the communication patterns among the corporate entities.



Figure II: The communication graph using the Yifan Hu layout. Each financial institution is assigned a colour so that communities are visible.

Various centrality measures, such as eigen centrality and betweenness centrality, were utilized to identify nodes with high influence on information flow within the graph. Unread emails were also taken into consideration during node selection. The emails associated with the selected nodes were flagged for further review by HCC employees, as they were likely to contain crucial information or contribute to the investigation.

3. Neural Network for Email Body Classification

A neural network model was employed for email body classification, which helped determine the relevance and importance of each email in the investigation. To prepare the email bodies for classification, text cleaning, and normalization techniques were applied. This ensured consistent and reliable input for the neural network model. A transformer ensemble model was employed to characterize the emails as 'Non-Applicable' (0) or 'Crucial' (1). The model leveraged true labels from emails previously read by HCC employees as part of the training data. By applying the ensemble model, emails were classified based on their relevance to the investigation.

By combining the outputs of the neural network model with the results from attachment analysis and graph analysis, a subset of suspicious emails emerged. The neural network's classification helped filter emails deemed crucial (labeled as '1'). Through the collective application of these techniques, the number of emails that had to be manually reviewed by HCC employees was significantly reduced allowing for a more efficient and targeted review process.

III. New workstreams

A. An AI-enabled documental retrieval system with semantic search

To streamline document retrieval and knowledge sharing, we plan to develop a vector database that utilizes a Large Language Model with Legal Expertise (LLM-LE) on top. This system will (i) perform semantic searches to retrieve semantically relevant documents, such as past cases and court decisions and (ii) potentially offer legal insights for an ongoing case based on the retrieved documents. By harnessing the power of NLP and machine learning, the HCC can enhance its research capabilities, access valuable precedents, and strengthen its ability to enforce competition laws.

B. Developing tools for social media and company monitoring

Monitoring social media serves multiple purposes for the Hellenic Competition Commission (HCC). Firstly, it aids in detecting potential antitrust violations by identifying suspicious activities that warrant further investigation. This includes monitoring social media posts, company updates, accounting reports, and investor updates for any indications of anti-competitive behavior. Secondly, social media monitoring helps in gathering evidence for antitrust cases. Additionally, social media monitoring allows the HCC to identify consumer concerns and gain insights into consumer sentiment. By monitoring social media conversations, the HCC can

understand consumer perspectives on various factors such as pricing, product quality, and access to goods and services. This information helps the HCC identify potential issues that may require intervention or further investigation to protect consumer interests and ensure fair competition. Lastly, social media monitoring plays a role in monitoring compliance with antitrust regulations. By monitoring social media posts and communications, the HCC can identify potential violations and initiate investigations accordingly.

C. An AI-revamp of the HCC's existing whistle-blower platforms

Integrating AI-based chatbots into the whistleblower reporting process offers several advantages. By posing specific questions, the AI-based chatbot can effectively guide whistleblowers, ensuring that they provide comprehensive information about alleged anti-competitive behavior. The chatbot's adaptability allows it to tailor its inquiries based on the whistleblower's responses, fostering a more interactive and personalized experience. This adaptive approach helps uncover additional details that may have been overlooked in a simple text submission, enhancing the quality of the information collected.

Moreover, AI-based chatbots possess the ability to seek clarifications and pose follow-up questions as necessary, ensuring the clarity and accuracy of the gathered information. Furthermore, chatbots exhibit multilingual proficiency, enabling them to comprehend and process diverse languages, even within the same sentence. This linguistic versatility facilitates communication with whistleblowers across various language backgrounds, accommodating their preferred means of expression. Additionally, leveraging the information provided by the whistleblower, AI-based chatbots can autonomously categorize and prioritize reports, eliminating the manual burden of tagging and archiving for HCC staff. This automated process streamlines the handling of whistleblower reports, allowing for more efficient and organized management.

D. AI-Assisted Web scraping

Generative AI can play a vital role in assisting with web scraping tasks, enabling the HCC to streamline the data collection and integration steps and add more sources of internet data. Models can be trained to understand the structure and content of web pages, facilitating the extraction of relevant data. By utilizing NLP, AI can accurately parse information from complex web pages, including textual data,

images, tables, and more. Additionally, AI can assist in ensuring the quality and integrity of scraped data.

E. Automatic Inquiry Assignment and Prioritization with NLP

The Directorate-General must, among other duties, assign inquiries to the relevant departments within the HCC. This is a time-consuming and error-prone process, which can be significantly reduced by leveraging AI to assign inquiries to their relevant Directorates and Units automatically. Models can be trained to understand the content and context of inquiries received by the HCC and extract key information. The subject matter, industry sector, and complexity of each inquiry can be determined, and it can be intelligently routed to the appropriate departments based on their expertise and workload. Finally, by considering factors such as caseload, available resources, specialization, historical data and user feedback, the HCC can continuously improve the model's performance in inquiry assignment.

IV. Conclusions

Integrating data science and data scientists in competition law enforcement and the day-to-day work of competition authorities presents one of the important challenges for competition law enforcers in the years to come. Advanced data analytics and generative AI may revolutionize the work of competition agencies and may enhance their capabilities to detect and punish anticompetitive activity. At the same time, these new tools offer significant capabilities in terms of market mapping and market monitoring, thus enhancing the ability of the competition authority to use a variety of tools (infringement decisions, settlements, sector inquiries, market investigation references, soft law approaches) and to make a deliberative and flexible (responsive) choice from the various regulatory strategies at their disposal. The Report presents the various workstreams of the HCC in-house Data Science team and explains how data science tools were employed in several competition law investigations.